Lab 9 • Numerical & Practice Worksheet

Functional Test Case Design | MCA Software Engineering

Q1. Boundary Value Testing

Assumption: A student may have from 0 to 5 books currently borrowed. The system must not allow an issue that would take the total above 5. Negative counts are invalid.

a) Classify these current borrowed-book counts as valid or invalid. b) For a request to issue one more book, write the expected system response.

Current books	Is current count valid?	Can one more book be issued?	Expected result / reason
-1			
0			
1			
4			
5			
6			

c) Write three boundary values around the borrowing limit: _____, _____, _____.

Q2. Equivalence Classes

Assumption: A valid member ID must contain exactly 6 digits (0-9). Letters and spaces are not accepted. Leading zero is allowed.

Input	Valid or invalid?	Reason / expected validation message
123456		
012345		
12345		
1234567		
12A456		

Name one valid class and two invalid classes: _____

Numerical Questions (continued)

Q3. Overdue Fine Calculation

Assumption: A book is due after 14 days. A fine of Rs. 2 is charged for each day beyond 14 days. For 14 days or fewer, the fine is Rs. 0.

Use: Overdue days = max(0, days kept - 14). Fine (Rs.) = 2 × overdue days.

Days kept	Overdue days (show work)	Fine (Rs.)	Expected system result
13			
14			
15			
18			
24			

Write one negative test for invalid input “days kept = -2”. Expected response: _____

Q4. Test Execution Percentages

Assumption: 42 cases were planned. During execution, 35 passed, 4 failed, and 3 were blocked (not executed).

a) Executed = Passed + Failed = _____ + _____ = _____ cases.

b) Pass rate among executed = Passed / Executed × 100 = _____ %.

c) Blocked rate among planned = Blocked / Planned × 100 = _____ %.

d) Why should blocked tests not be counted as passed? _____

Q5. Requirement Coverage

Assumption: The project contains six functional requirements: R-01 to R-06. The tests cover R-01, R-02, R-03, R-04, and R-06.

a) Number of covered requirements = _____. b) Uncovered requirement = _____.

c) Requirement coverage = (Covered requirements / Total requirements) × 100 = _____ %.

d) Write one new test-case scenario for the uncovered requirement: _____

Practice: Create Your Own Test Cases

Q6. Short Practical Case Study

Assumption: In a sample Library Management System, users can search for a book by title. If the book exists, its title and availability are shown. If no match exists, the system shows “No books found.”

Create three test cases: one normal valid search, one no-match search, and one empty-input search. For empty input, state your proposed expected behavior clearly and label it as an assumption if the SRS does not define it.

Test case ID	Test input	Positive / negative	Expected result
TC-01			
TC-02			
TC-03			

Mini reflection

Explain in one sentence why writing “system works correctly” is not a good expected result:

_____

_____

Note: For students who use a different project, replace the sample limits and rules with the requirements recorded in their SRS.